Pre-Lab: Empirical Formula of Magnesium Oxide

Name: _____ Date: _____ Period: _____

Complete before lab day. In lab you burn a piece of magnesium ribbon and use the masses to find the mole ratio of Mg to O. Show your work and use units.

Part A: Background

1. What is an empirical formula? Why can't we find it by just looking at the compound?

2. Mg forms Mg^{2+} and O forms O^{2-}. Predict the formula of magnesium oxide and the mole ratio of Mg to O.

3. Balance: ____ Mg + ____ O_2 → ____ MgO

4. Magnesium is so reactive that it also reacts with the nitrogen in air, making magnesium nitride. Write the formula of magnesium nitride (Mg^{2+} and N^{3-}) and balance: ____ Mg + ____ N_2 → ____ Mg_3N_2

5. After the first heating you add water. Balance the reaction of the nitride with water, and the reaction when $Mg(OH)_2$ is heated. Then explain how the water step helps the experiment.
____ Mg_3N_2 + ____ H_2O → ____ $Mg(OH)_2$ + ____ NH_3 ____ $Mg(OH)_2$ → ____ MgO + ____ H_2O

6. Why is the lid left slightly ajar during heating, instead of fully off or fully on?

7. You cannot weigh the oxygen that reacts. Explain how you will find the mass of oxygen from three masses.

Part B: Practice calculation and precision

8. In a trial, 0.1500 g of Mg burns to give 0.2487 g of product. Find the mass of oxygen, the moles of each element, and the empirical formula. Show every step.

9. The balance reads 25.4310 g. How many decimal places is that? When you subtract two such masses, how many decimal places belong in the answer? Why is the last digit allowed to drift on a warm crucible?

10. List three safety precautions for this lab and the hazard each prevents.

11. Predict. How would you prove that all the water is gone and the mass is final? (Hint: the idea of constant mass.)

Lab: Empirical Formula of Magnesium Oxide (47 minutes)

Name: _____ Date: _____ Period: _____ Partner: _____

Purpose: Find the empirical formula of magnesium oxide by burning a weighed piece of Mg, finding the mass of oxygen that combined by difference, and comparing the mole ratio with the formula you predicted in the pre-lab.

SAFETY

- Goggles on all period. Hair tied back. Do not look directly at burning magnesium.
- Tongs only for the crucible and lid. Set hot items on the ceramic pad, never on the bench.
- Add water slowly to the cooled ash, never to a glowing crucible. Waft the ammonia smell, do not sniff.

Materials: numbered crucible and lid (mass on label), pre-massed Mg ribbon (mass on bag), ring stand with ring and clay triangle, Bunsen burner, tongs, ceramic pad, dropper bottle of distilled water, 0.0001 g balance (shared).

Procedure (minutes are a guide)

1. **(0–4)** Goggles on. Set the clay triangle on the ring so the crucible sits in it. Record the **crucible + lid mass** and the **Mg mass** from the labels. Do not weigh them again.
2. **(4–6)** Coil the Mg ribbon in the crucible. Set the lid on **slightly ajar** (a small gap, not off). Set the crucible in the triangle.
3. **(6–16)** Light the burner. Heat gently for 2 minutes, then strongly. When the Mg ignites, keep the lid ajar. **Do not stare at the flame.** Heat until the ribbon is fully burned and no longer glowing (about 8 minutes). Record what you see.
4. **(16–19)** Turn the burner off. Use tongs to set the crucible on the ceramic pad. Cool 3 minutes. Start the calculation setup.
5. **(19–21)** The crucible must no longer glow. Remove the lid with tongs and add about **10 drops of distilled water** slowly onto the ash. Waft any odour toward you. Record the odour and colour.
6. **(21–28)** Replace the lid slightly ajar and heat gently for 2 minutes to drive off the water, then strongly for 4 to 5 minutes. Turn off the burner.
7. **(28–31)** Cool 3 minutes on the pad. Sign up for a balance.
8. **(31–36)** Mass the crucible, lid and product together on the balance (4 decimal places). The last digit may drift: record the first reading that holds for 3 seconds. Carry the crucible with tongs.
9. **(36–47)** Finish the calculations, add your results to the class table, then clean up: let equipment cool, return crucible and tongs, wipe your area, wash hands. The product goes in the trash.

Data

	Mass (g)
Mass of crucible + lid (from label)	
Mass of Mg ribbon (from bag)	
Mass of crucible + lid + product (after heating)	
Mass of product, MgO (final − crucible + lid)	
Mass of oxygen (product − Mg)	

Observations (before heating, during burning, after adding water, final product):

Calculations (show setup and units; your last digit is limited by the balance)

	Work	Answer
Moles of Mg (÷ 24.31 g/mol)		
Moles of O (÷ 16.00 g/mol)		
Mole ratio O : Mg (divide both by the smaller)		
Empirical formula (whole numbers)		
Percent error in the O : Mg ratio (accepted 1.00)		

Class data (record every group)

Group	Mg (g)	MgO (g)	O (g)	O : Mg ratio
Class average				

Analysis questions

1. State your experimental O : Mg mole ratio and empirical formula. Does it agree with your pre-lab prediction? Use the class average in your answer.

2. Give two observations from the lab that support the formation of an oxide (or a nitride that was converted to an oxide).

3. The ribbon gained mass. Where did the extra mass come from? Which law does this show?

4. Name one source of error that could make your ratio too high and one that could make it too low. Explain each.

5. How does the water step improve your result? What would the ratio look like without it?

Extension questions (honors)

E1. Theoretical yield. Using your Mg mass, calculate the mass of MgO you should have made and your percent yield.

E2. Percent composition. Find the percent Mg by mass in your product and compare it with the percent Mg in MgO.

E3. Gas volume. How many liters of O_2 gas at STP reacted with your Mg? (Hint: moles O_2 = moles O ÷ 2.)

E4. Suppose all of the Mg had formed Mg_3N_2 and you did the calculation as if the extra mass were oxygen. What O : Mg mole ratio would you get? What does that tell you about the effect of leaving the nitride in the product?

E5. Question the evidence. Hot magnesium can also react with carbon dioxide: $2\ Mg + CO_2 \rightarrow 2\ MgO + C$. A student's product is grey-black. What does the colour suggest and how would the carbon change the O : Mg ratio?

E6. A student burns 0.2698 g of aluminum and gets 0.5098 g of aluminum oxide. Find the empirical formula (Al = 26.98 g/mol). Hint: the ratio may not be 1 : 1.